

← Main menu

Build and Deploy Machine Learning Solutions on Vertex AI

Course - 7 ✓ Complete hours

🏠 Course overview

Build and Deploy Machine Learning Solutions on Vertex AI

Identify Damaged Car Parts with Vertex AutoML Vision

Deploy a BigQuery ML Customer Churn Classifier to Vertex AI for Online Predictions

Vertex AI Pipelines: Gwik Start

Build and Deploy Machine Learning Solutions on Vertex AI: Challenge Lab

Your Next Steps

Course > Build and Deploy Machine Learning Solutions on Vertex AI >

Lab setup instructions and requirements

End Lab

01:28:46

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud console

Username

student-b2-e1246bf28d81f

Password

3tauXMuon1cX

Project ID

qwiklabs-gcp-02-a7f4c96f

Identify Damaged Car Parts with Vertex AutoML Vision

🏷️ Lab ⌚ 1 hour 30 minutes 💰 No cost 🏆 Intermediate

★★★★★ Rate Lab

ⓘ This lab may incorporate AI tools to support your learning.

Lab instructions and tasks

GSP972

Overview

Objectives

Setup and requirements

Task 1. Upload training Images to Cloud Storage

Task 2. Create a dataset

Task 3. Inspect images

Task 4. Train your model

Task 5. Request a prediction from a hosted model

Congratulations!

0/100

Identify Damaged Car Parts with Vertex AutoML Vision

🏷️ Lab ⌚ 1 hour 30 minutes 💰 No cost 🏆 Intermediate

★★★★★ Rate Lab

ⓘ This lab may incorporate AI tools to support your learning.

GSP972



Overview

Vertex AI brings together the Google Cloud services for building ML under one unified interface. As a result, you can train, deploy, and manage models using [Vertex AI](#) or custom code training and all your models are stored in one central model repository. These models can now be deployed to the same endpoints on Vertex AI.

AutoML Vision helps anyone with limited Machine Learning (ML) expertise train high quality image classification models. In this hands-on lab, you will learn how to produce a custom ML model that automatically recognizes damaged car parts. Since the time it takes to train the model is above the time limit of the lab, you will interact and request predictions from a hosted model in a different project trained on the same dataset. You will then tweak the values of the data for the prediction request and examine how it changes the resulting prediction from the model.

Objectives

In this lab, you learn how to:

- Upload a labeled dataset to Cloud Storage using a CSV file and connect it to Vertex AI as a Managed Dataset
- Inspect uploaded images to ensure there are no errors in your dataset
- Kick off an AutoML Vision model training job
- Request predictions from a hosted model trained on the same dataset

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

Note: Use only the student account for this lab. If you use a different Google Cloud

How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. If you need to pay for the lab, a dialog opens for you to select your payment method. On the left is the Lab Details pane with the following:

- The Open Google Cloud console button
- Time remaining
- The temporary credentials that you must use for this lab
- Other information, if needed, to step through this lab

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

manage them by deleting the entire history.

The lab spins up resources, and then opens another tab that shows the Sign in page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

student-02-e1246bf28d81@qwiklabs.net

You can also find the Username in the Lab Details pane.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

3tauXMuo1cX

You can also find the Password in the Lab Details pane.

6. Click **Next**.

Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.

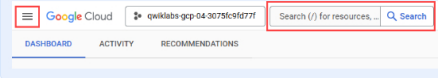
Note: Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To access Google Cloud products and services, click the **Navigation menu** or type the service or product name in the **Search** field.



Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** at the top of the Google Cloud console.

2. Click through the following windows:

- Continue through the Cloud Shell information window.
- Authorize Cloud Shell to use your credentials to make Google Cloud API calls.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `qwiklabs-gcp-02-a7f4c966bc9d`. The output contains a line that declares the **Project_ID** for this session:

```
Your Cloud Platform project in this session is set to qwiklabs-gcp-02-a7f4c966bc9d
```

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

3. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

4. Click **Authorize**.

Output:

```
ACTIVE: *
ACCOUNT: student-02-e1246bf28d81@qwiklabs.net

To set the active account, run:
$ gcloud config set account 'ACCOUNT'
```

5. (Optional) You can list the project ID with this command:

```
gcloud config list project
```

Output:

```
[core]
project = qwiklabs-gcp-02-a7f4c966bc9d
```

Note: For full documentation of `gcloud`, in Google Cloud, refer to [the gcloud CLI overview guide](#).

Task 1. Upload training images to Cloud Storage

In this task you will upload the training images you want to use to Cloud Storage. This will make it easier to import the data into Vertex AI later.

To train a model to classify images of damaged car parts, you need to provide the machine with labeled training data. The model will use the data to develop an understanding of each image, differentiating between car parts and those with damages on them.

Note: For the purposes of this lab, you won't need to label images because a labeled dataset (i.e. image plus label) in a CSV file has been provided. The next section outlines the steps to use the CSV file.

In this example, your model will learn to classify five different damaged car parts: **bumper**, **engine compartment**, **hood**, **lateral**, and **windshield**.

Create a Cloud Storage bucket

1. To start, open a new Cloud Shell window and execute the following commands to set some environment variables:

```
export PROJECT_ID=$DEVSHLL_PROJECT_ID
export BUCKET=$PROJECT_ID
```

2. Next, to create a Cloud Storage bucket, execute the following command:

```
gsutil mb -p $PROJECT_ID \
-c standard \
-l "us-central1" \
gs://${BUCKET}
```

Upload car images to your Storage Bucket

The training images are publicly available in a Cloud Storage bucket. Again, copy and paste the script template below into Cloud Shell to copy the images into your own bucket.

1. To copy images into your Cloud Storage bucket, execute the following command:

```
gsutil -m cp -r gs://car_damage_lab_images/* gs://${BUCKET}
```

2. In the navigation pane, click **Cloud Storage > Buckets**.
3. Click the **Refresh** button at the top of the Cloud Storage browser.
4. Click on your bucket name. You should see five folders of photos for each of the five different damaged car parts to be classified:

Buckets > `qwiklabs-gcp-01-b62a81e8986-vcn`

[UPLOAD FILES](#) [UPLOAD FOLDER](#) [CREATE FOLDER](#) [MANAGE HOLDS](#) [DOWNLOAD](#) [DELETE](#)

Filter by name prefix only Filter objects and folders

☐	Name	Size	Type	Created	Storage class	Last modified
☐	bumper/	--	Folder	--	--	--
☐	engine_compartment/	--	Folder	--	--	--
☐	hood/	--	Folder	--	--	--
☐	lateral/	--	Folder	--	--	--
☐	windshield/	--	Folder	--	--	--

5. Optionally, you can click one of the folders and check out the images inside.

Check that car images are now organized and ready for training.

Click *Check my progress* to verify the objective.

☐

Upload car images to your Storage Bucket

[Check my progress](#)

Task 2. Create a dataset

In this task, you create a new dataset and connect your dataset to your training images to allow Vertex AI to access them.

Normally, you would create a CSV file where each row contains a URL to a training image and the associated label for that image. In this case, the CSV file has been created for you; you just need to update it with your bucket name and upload the CSV file to your Cloud Storage bucket.

Update the CSV file

Copy and paste the script templates below into Cloud Shell and press enter to update,

1. To create a copy of the file, execute the following command:

```
gsutil cp gs://car_damage_lab_metadata/data.csv .
```

2. To update the CSV with the path to your storage, execute the following command:

```
sed -i -e "s/car_damage_lab_images/${BUCKET}/g" ./data.csv
```

3. Verify your bucket name was inserted into the CSV properly:

```
cat ./data.csv
```

4. To upload the CSV file to your Cloud Storage bucket, execute the following command:

```
gsutil cp ./data.csv gs://${BUCKET}
```

5. Once the command completes, click the **Refresh** button at the top of the Cloud Storage browser and open your bucket.

6. Confirm that the `data.csv` file is listed in your bucket.

Buckets > `gs://car-damage-lab-images`

UPLOAD FILES UPLOAD FOLDER CREATE FOLDER MANAGE HOLDS DOWNLOAD DELETE

Filter by name prefix only ▾ Filter Filter objects and folders

☐	Name	Size	Type	Created	Storage class	Last modified
☐	bumper/	—	Folder	—	—	—
☒	data.csv	7.9 KB	text/csv	Sep 29, 2021, 1:41:46 PM	Standard	Sep 29, 2021, 1:41:46 PM
☐	engine_compartment/	—	Folder	—	—	—
☐	hood/	—	Folder	—	—	—
☐	lateral/	—	Folder	—	—	—
☐	windshield/	—	Folder	—	—	—

Create a managed dataset

1. In the Google Cloud Console, on the **Navigation menu** (☰) click **Vertex AI > Dashboard**.
2. Click **Enable All Recommended APIs** if it is not already enabled.
3. From the Vertex AI navigation menu on the left, click **Datasets**.
4. At the top of the console, click **+ Create**.

6. Select **Single-label classification**. (Note: in your own projects, you may want to check the "Multi-label Classification" box if you're doing [multi-class classification](#)).
7. Select the Region as `us-central1`.
8. Click **Create**.

Connect your dataset to your training images

In this section, you will choose the location of your training images that you uploaded in the previous step.

1. In the **Select an import method** section, click **Select import files from Cloud Storage**.
2. In the **Select import files from Cloud Storage** section, click **Browse**.
3. Follow the prompts to navigate to your storage bucket and click your `data.csv` file. Click **Select**.
4. Once you've properly selected your file, a green checkbox appears to the left of the file path. Click **Continue** to proceed.

Note: It will take around 9 to 12 minutes for your images to import and be aligned with their categories. You'll need to wait for this step to complete before checking your progress.

5. Once the import has completed, prepare for the next section by clicking the **Browse** tab. (Hint: You may need to refresh the page to confirm.)

Click *Check my progress* to verify the objective.

Create a dataset

Check my progress

Task 3. Inspect images

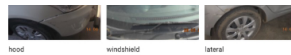
In this task, you examine the images to ensure there are no errors in your dataset.

IMPORT	BROWSE	ANALYZE
All100		Filter Filter items
Labelled100		
training	100	
Validation	20	
Test	15	
Filter Filter labels +		
bumper	20	
engine_compartment	20	
hood	20	
lateral	20	
windshield	20	
ADD NEW LABEL		



windshield bumper windshield



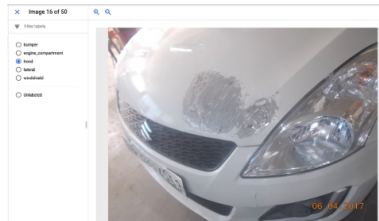


Check image labels

1. If your browser page has refreshed, click **Datasets**, select your image name, and then click **Browse**.
2. Under **Filter labels**, click any one of the labels to view the specific training images. (Example: *engine_compartment*.)

Note: If you are building a predictive model, you'd want at least 100 images per label to ensure high accuracy. One option is to use only 20 images of each type were used so the model could train quickly.

3. If an image is labeled incorrectly, you can click on it to select the correct label or delete the image from your training set:



4. Next, click on the **Analyze** tab to view the number of images per label. The **Label Stats** window appears on your browser.

Task 4. Train your model

You're ready to start training your model! Vertex AI handles this for you automatically, without requiring you to write any of the model code.

1. From the right-hand side, click **Train New Model**.
2. From the **Training method** window, leave the default configurations and select **AutoML** as the training method. Click **Continue**.
3. From the **Model details** window, enter a name for your model, use: `damaged_car_parts_model1`. Click **Continue**.
4. From the **Training options** window, select **Higher accuracy (new)** and click **Continue**.
5. From **Compute and pricing** window, set your budget to **8** maximum node hours.
6. Click **Start Training**.

Note: Model training can take longer than the allotted time to complete the lab. The model **does not need to finish training** to continue to the next section.

Click *Check my progress* to verify the objective.

Train your model

Task 5. Request a prediction from a hosted model

For the purposes of this lab, a model trained on the exact same dataset is hosted in a different project so that you can request predictions from it while your local model finishes training, as it is likely that the local model training will exceed the limit of this lab.

A proxy to the pre-trained model is set up for you so you don't need to run through any extra steps to get it working within your lab environment.

To request predictions from the model, you will send predictions to an endpoint inside of your project that will forward the request to the hosted model and return back the output. Sending a prediction to the AutoML Proxy is very similar to the way that you would interact with your model you just created, so you can use this as practice.

Get the name of AutoML proxy endpoint

1. In the Google Cloud Console, on the **Navigation menu**, click **Cloud Run**.
2. Click **automl-proxy**.

SERVICES					
Filter services					
☐	Name ↑	Req/sec	Region	Authentication	Ingress
☒	automl-proxy	0	us-central1	Allow unauthenticated	All

3. Copy the URL to the endpoint. It should look something like: `https://automl-proxy-xfpm6c62ta-uc.a.run.app`.

automl-proxy Region: us-central1 URL: <https://automl-proxy-xfpm6c62ta-uc.a.run.app>

You will use this endpoint for the prediction request in the next section.

Create a prediction request

- On the Cloud Shell toolbar, click **Open Editor**. If prompted click **Open in New Window**.
- Click **File > New File**.
- Copy the following content into the new file you just created:

```
{
  "instances": [ {
    "content":
"/9j/4AAQSkZJRgABAQAAQABAAQ/4gIoSUNDX1BST0ZJTEUAAQAAAAIYAAAAQwAJ
"
```

- Click **File > Save** then select your path from dropdown (/home/student_xx_XXXX).
- Name your file as `payload.json` and then click **Save**.

For reference, the content you supplied is a Base64 string from the following image.



- Next, set the following environment variables. Copy in your AutoML Proxy URL you retrieved in earlier.

```
AUTOML_PROXY=<automl-proxy url>
INPUT_DATA_FILE=payload.json
```

- Perform a API request to the AutoML Proxy endpoint to request the prediction from the hosted model:

```
curl -X POST -H "Content-Type: application/json"
$AUTOML_PROXY/v1 -d "@$(INPUT_DATA_FILE)"
```

If you ran a successful prediction, your output should resemble the following:

```
{
  "predictions": [ {
    "confidences": [ 0.951557755 ],
    "displayNames": [ "bumper" ],
    "id":

```

For this model, the prediction results are pretty self-explanatory. The `displayNames` field should correctly predict a bumper with a high confidence threshold. Now, you can change the Base64 encoded image value in the JSON file you created.

Click *Check my progress* to verify the objective.

☐ Create the prediction request
 ☒ **Check my progress**

- Right-click on each image below, then select **Save image As....**
- Follow the prompts to save each image with a unique name. (*Hint: Assign a simple name like 'Image1' and 'Image2' to assist with uploading.*)



11. Open the [Base64 Image Encoder](#) follow the instructions to upload and encode an image to a Base64 string.

12. Replace the Base64 encoded string value in the `content` field in your JSON payload file, and run the prediction again. Repeat for the other image(s).

How did your model do? Did it predict all three images correctly? You should see the following outputs, respectively:

```
{
  "predictions": [
    {
      "ids": [
        "5419751198548431368"
      ],
      "confidences": [
        0.985487759
      ]
    }
  ]
}
```

```
{
  "predictions": [
    {
      "displayNames": [
        "hood"
      ],
      "ids": [
        "3113988189326737488"
      ],
      "confidences": [
        0.985487759
      ]
    }
  ]
}
```

Congratulations!

In this lab, you learned how to train your own custom machine learning model and generate predictions on hosted model via an API request. You uploaded training images to Cloud Storage and used a CSV file for Vertex AI to find these images. You inspected the labeled images for any discrepancies before finally evaluating a trained model. Now you've got what it takes to train a model on your own image dataset!

Next steps / learn more

- Read more about Vertex AI in the [Vertex AI Documentation](#)

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. [Our classes](#) include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. [Certifications](#) help you validate and prove your skill and expertise in Google Cloud technologies.

Manual Last Updated November 29, 2024

Lab Last Tested November 29, 2024

Copyright 2025 Google LLC. All rights reserved. Google and the Google logo are trademarks of Google LLC. All other company and product names may be trademarks of the respective companies with which they are associated.